ECONOMICS Week 05 Demand, Supply, and Market Equilibrium

Sardar Shahbaz A. Naqvi

Outline

- Firms and Households: The Basic Decision-Making Units
- Input Markets and Output Markets: The Circular Flow
- Demand in Product/Output Markets
- Supply in Product/Output Markets
- Market Equilibrium
- Demand and Supply in Product Markets: A Review
- Looking Ahead: Markets and the
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Firms and Households: The Basic Decision-Making Units

firm An organization that transforms resources (inputs) into products (outputs). Firms are the primary producing units in a market economy.

entrepreneur A person who organizes, manages, and assumes the risks of a firm, taking a new idea or a new product and turning it into a successful business.

households The consuming units in an economy.

Input Markets and Output Markets: The Circular Flow

product or output markets The markets in which goods and services are exchanged.

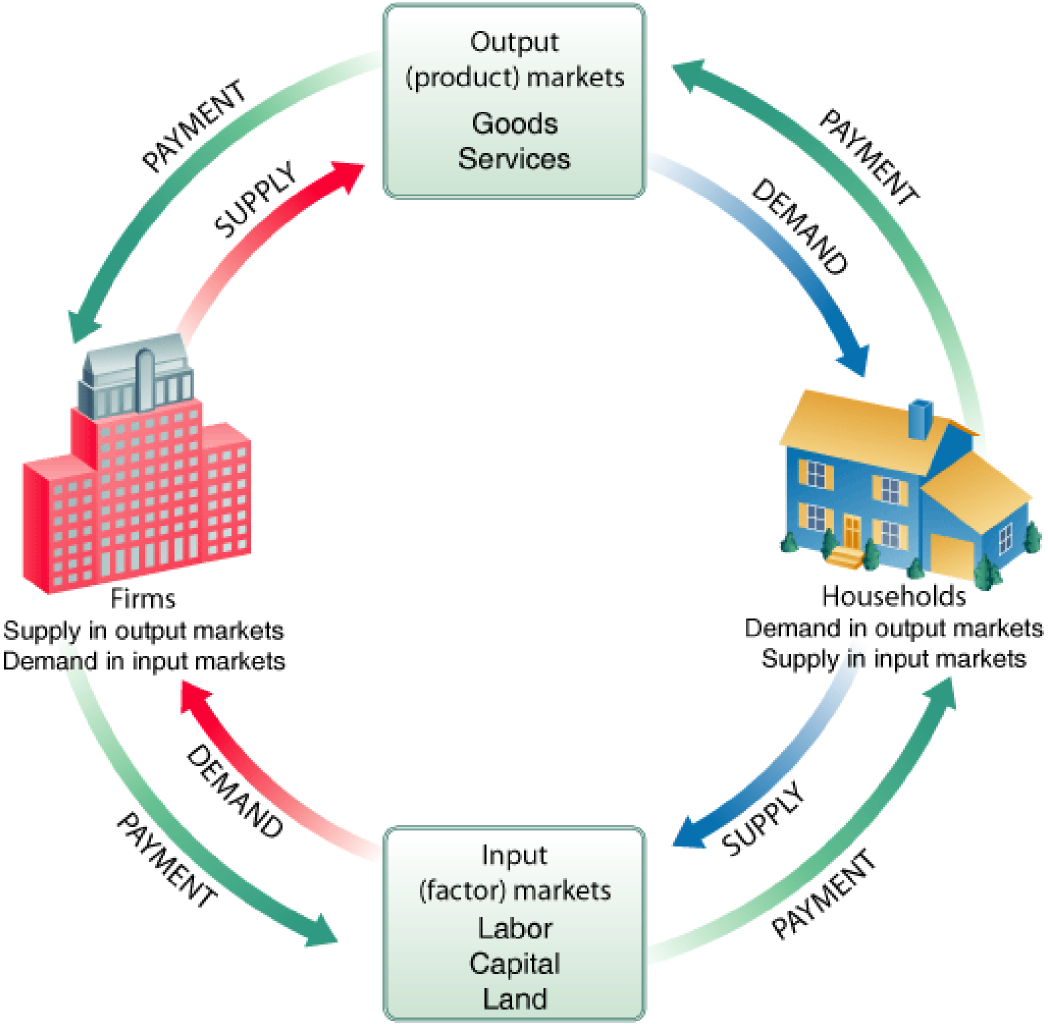
input or factor markets The markets in which the resources used to produce products are exchanged.

Input Markets and Output Markets: The Circular Flow

FIGURE 3.1 The Circular Flow of Economic Activity

Diagrams like this one show the circular flow of economic activity, hence the name circular flow diagram. Here goods and services flow clockwise: Labor services supplied by household's flow to firms, and goods and services produced by firms flow to households.

Payment (usually money) flows in the opposite (counterclockwise) direction: Payment for goods and services flows from households to firms, and payment for labor services flows from firms to households.



Input Markets and Output Markets: The Circular Flow

labor market The input/factor market in which households supply work for wages to firms that demand labor.

capital market The input/factor market in which households supply their savings, for interest or for claims to future profits, to firms that demand funds to buy capital goods.

Input Markets and Output Markets: The Circular Flow

land market The input/factor market in which households supply land or other real property in exchange for rent.

factors of production The inputs into the production process. Land, labor, and capital are the three key factors of production.

Input and output markets are connected through the behavior of both firms and households. Firms determine the quantities and character of outputs produced and the types and quantities of inputs demanded. Households determine the types and quantities of products demanded and the quantities and types of inputs supplied.

Demand in Product/Output Markets

A household's decision about what quantity of a particular output, or product, to demand depends on a number of factors, including:

- The *price of the product* in question.
- The *income available* to the household.
- The household's *amount of accumulated wealth.*
- The *prices of other products* available to the household.
- The household's *tastes and preferences.*
- The household's *expectations* about future income, wealth, and prices.

Demand in Product/Output Markets

quantity demanded The amount (number of units) of a product that a household would buy in a given period if it could buy all it wanted at the current market price.

Demand in Product/Output Markets

Changes in Quantity Demanded versus Changes in Demand

The most important relationship in individual markets is that between market price and quantity demanded.

Changes in the price of a product affect the *quantity demanded* per period. Changes in any other factor, such as income or preferences, affect *demand*. Thus, we say that an increase in the price of Coca-Cola is likely to cause a decrease in the *quantity of Coca-Cola demanded*. However, we say that an increase in income is likely to cause an increase in the *demand* for most goods.

Demand in Product/Output Markets

Price and Quantity Demanded: The Law of Demand

demand schedule A table showing how much of a given product a household would be willing to buy at different prices.

demand curve A graph illustrating how much of a given product a household would be willing to buy at different prices.

Demand in Product/Output Markets

Price and Quantity Demanded: The Law of Demand

TABLE 3.1 Anna's Demand Schedule for Telephone Calls

Price (Per Call)	Quantity Demanded (Calls Per Month)
$ 0	30
.50	25
3.50	7
7.00	3
10.00	1
15.00	0

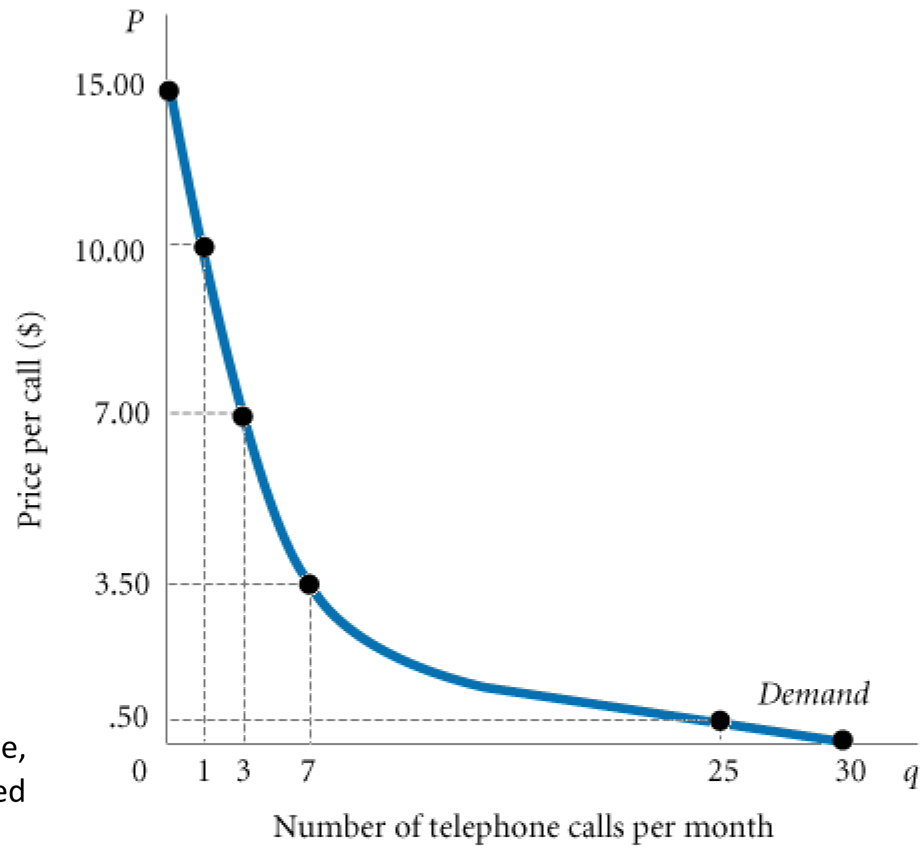


FIGURE 3.2 Anna's Demand Curve

The relationship between price (*P*) and quantity demanded (*q*) presented graphically is called a demand curve. Demand curves have a negative slope, indicating that lower prices cause quantity demanded to increase. Note that Anna's demand curve is blue; demand in product markets is determined by household choice.

Demand in Product/Output Markets

Price and Quantity Demanded: The Law of Demand

Demand Curves Slope Downward

law of demand The negative relationship between price and quantity demanded: As price rises, quantity demanded decreases; as price falls, quantity demanded increases.

It is reasonable to expect quantity demanded to fall when price rises, *ceteris paribus*, and to expect quantity demanded to rise when price falls, *ceteris paribus*. Demand curves have a negative slope.

Demand in Product/Output Markets

Price and Quantity Demanded: The Law of Demand

Other Properties of Demand Curves

Two additional things are notable about Anna's demand curve.

As long as households have limited incomes and wealth, all demand curves will intersect the price axis. For any commodity, there is always a price above which a household will not or cannot pay. Even if the good or service is very important, all households are ultimately constrained, or limited, by income and wealth.

That demand curves intersect the quantity axis is a matter of common sense. Demand in a given period of time is limited, if only by time, even at a zero price.

Demand in Product/Output Markets

Price and Quantity Demanded: The Law of Demand

Other Properties of Demand Curves

To summarize what we know about the shape of demand curves:

1. They have a negative slope. An increase in price is likely to lead to a decrease in quantity demanded, and a decrease in price is likely to lead to an increase in quantity demanded.
2. They intersect the quantity (*X*-) axis, a result of time limitations and diminishing marginal utility.
3. They intersect the price (*Y*-) axis, a result of limited incomes and wealth.

Demand in Product/Output Markets

Other Determinants of Household Demand

Income And Wealth

income The sum of all a household's wages, salaries, profits, interest payments, rents, and other forms of earnings in a given period of time. It is a flow measure.

wealth *or* net worth The total value of what a household owns minus what it owes. It is a stock measure.

Demand in Product/Output Markets

Other Determinants of Household Demand

Income And Wealth

normal goods Goods for which demand goes up when income is higher and for which demand goes down when income is lower.

inferior goods Goods for which demand tends to fall when income rises.

Demand in Product/Output Markets

Other Determinants of Household Demand

Prices of Other Goods and Services

substitutes Goods that can serve as replacements for one another; when the price of one increases, demand for the other increases.

perfect substitutes Identical products.

complements, complementary goods Goods that "go together"; a decrease in the price of one results in an increase in demand for the other and vice versa.

Demand in Product/Output Markets

Other Determinants of Household Demand

Tastes and Preferences

Income, wealth, and prices of goods available are the three factors that determine the combinations of goods and services that a household is *able* to buy.

Changes in preferences can and do manifest themselves in market behavior.

Within the constraints of prices and incomes, preference shapes the demand curve, but it is difficult to generalize about tastes and preferences. First, they are volatile. Second, tastes are idiosyncratic.

Demand in Product/Output Markets

Other Determinants of Household Demand

Expectations

What you decide to buy today certainly depends on today's prices and your current income and wealth.

There are many examples of the ways expectations affect demand.

Increasingly, economic theory has come to recognize the importance of expectations.

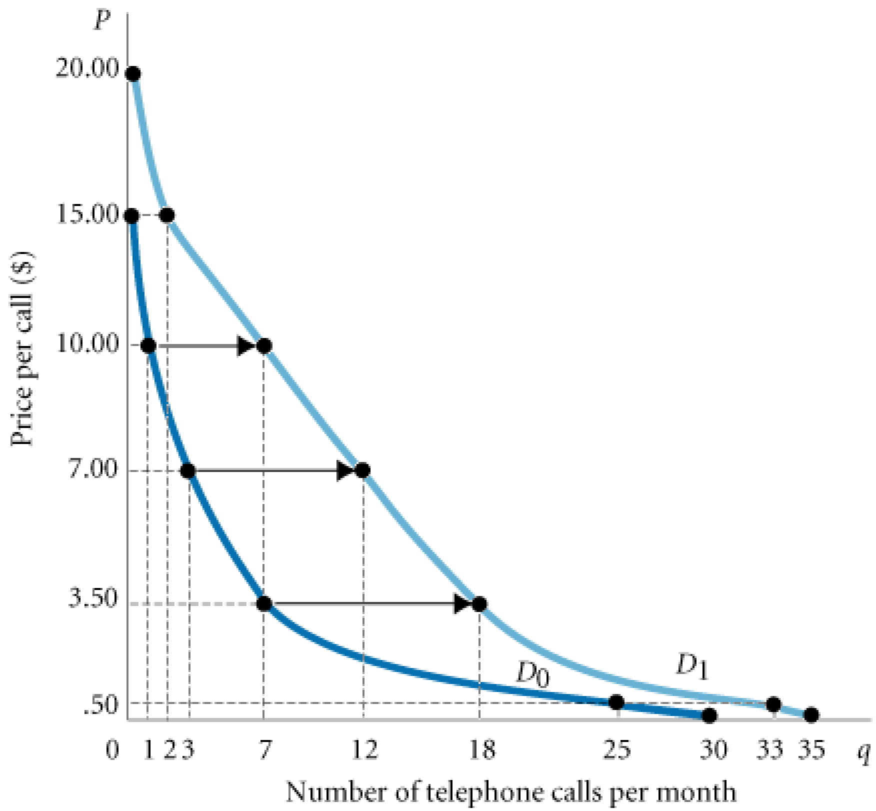
It is important to understand that demand depends on more than just *current* incomes, prices, and tastes.

Demand in Product/Output Markets

Shift of Demand versus Movement Along a Demand Curve

TABLE 3.2 Shift of Anna's Demand Schedule Due to increase in Income

	Schedule D_0	Schedule D_1
Price (Per Call)	**Quantity Demanded (Calls Per Month at an Income of $300 Per Month)**	**Quantity Demanded (Calls Per Month at an Income of $600 Per Month)**
$ 0.00	30	35
0.50	25	33
3.50	7	18
7.00	3	12
10.00	1	7
15.00	0	2
20.00	0	0



□ FIGURE 3.3 Shift of a Demand Curve Following a Rise in Income

When the price of a good changes, we move *along* the demand curve for that good.
When any other factor that influences demand changes (income, tastes, and so on), the relationship between price and quantity is different; there is a *shift* of the demand curve, in this case from D_0 to D_1. Telephone calls are normal goods.

Demand in Product/Output Markets

Shift of Demand versus Movement Along a Demand Curve

shift of a demand curve The change that takes place in a demand curve corresponding to a new relationship between quantity demanded of a good and price of that good. The shift is brought about by a change in the original conditions.

movement along a demand curve The change in quantity demanded brought about by a change in price.

Change in price of a good or service leads to
→ Change in *quantity demanded* (movement along the demand curve).

Change in income, preferences, or prices of other goods or services leads to
→ Change in *demand* (shift of the demand curve).

Demand in Product/Output Markets

Shift of Demand versus Movement Along a Demand Curve

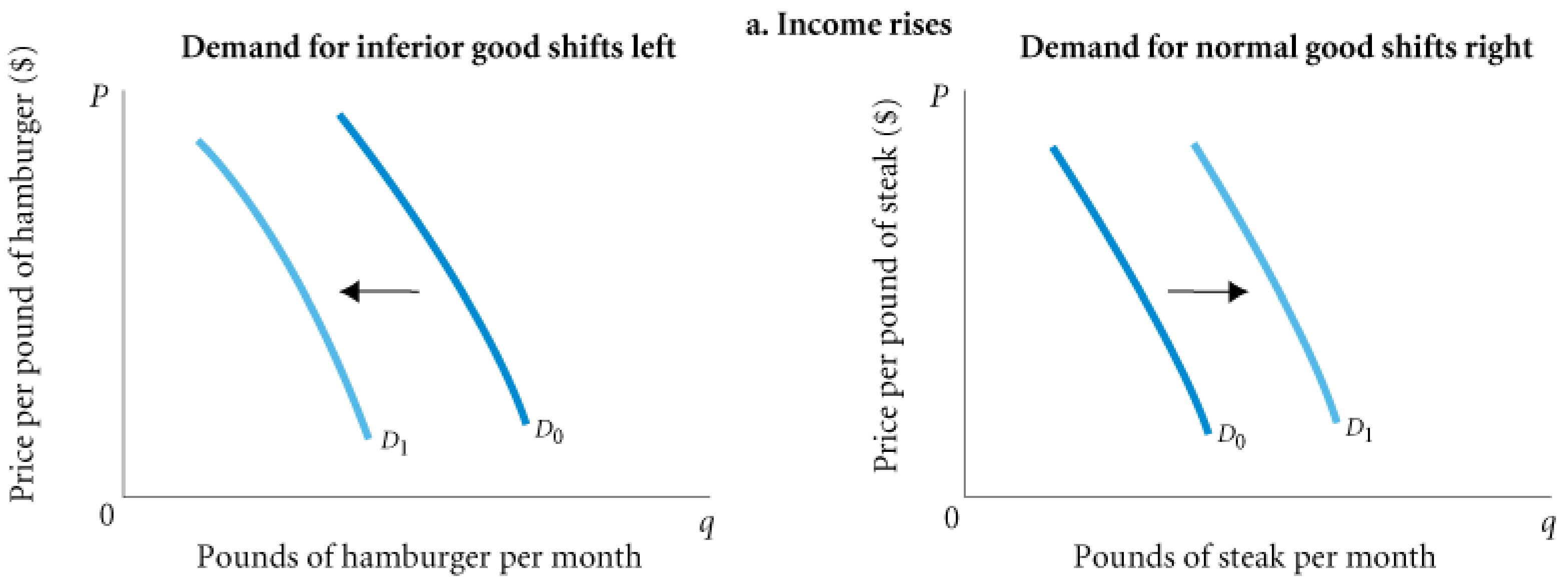


 FIGURE 3.4 Shifts versus Movement Along a Demand Curve

a. When income increases, the demand for inferior goods shifts to the left and the demand for normal goods shifts to the right.

Demand in Product/Output Markets

Shift of Demand versus Movement Along a Demand Curve

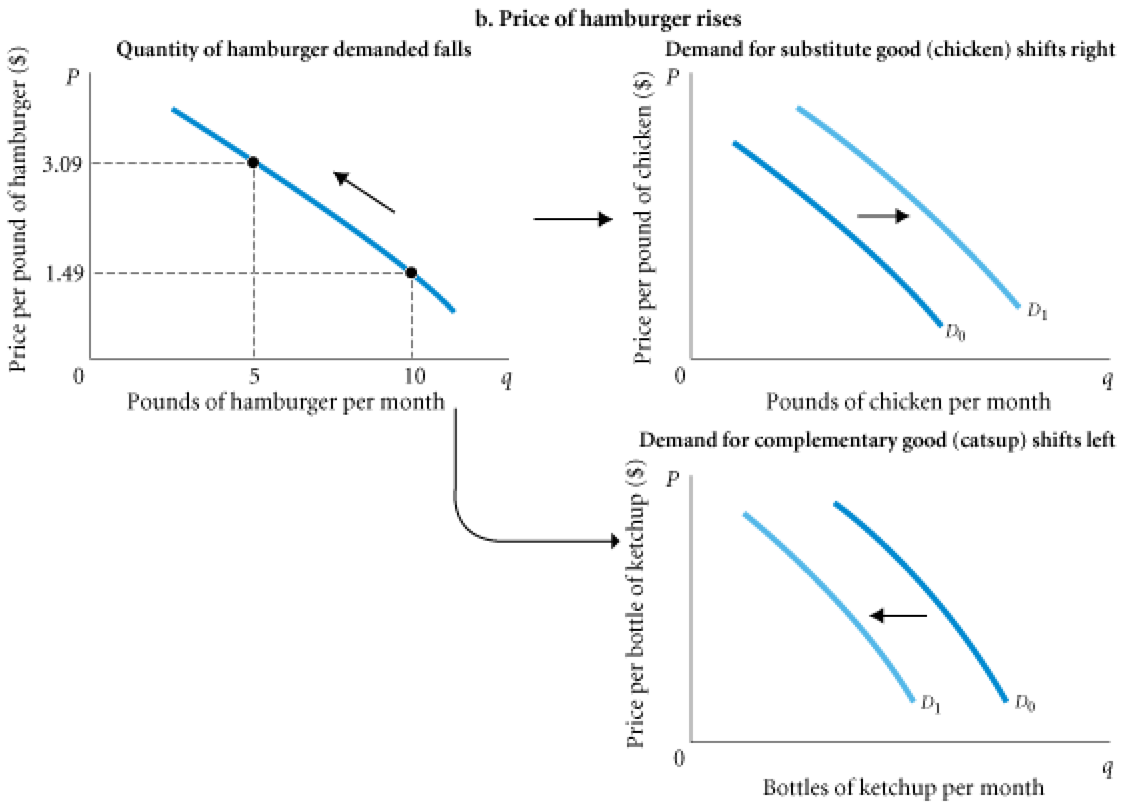


 FIGURE 3.4 Shifts versus Movement Along a Demand Curve (continued)

b. If the price of hamburger rises, the quantity of hamburger demanded declines— his is a movement along the demand curve.
The same price rise for hamburger would shift the demand for chicken (a substitute for hamburger) to the right and the demand for ketchup (a complement to hamburger) to the left.

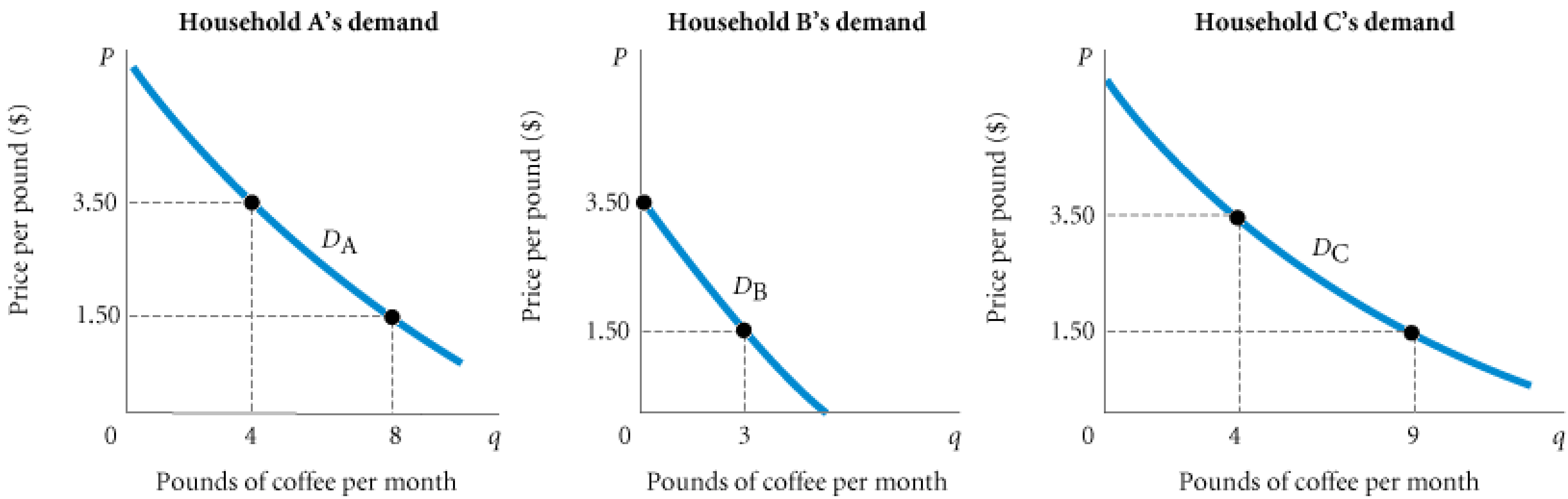
Demand in Product/Output Markets

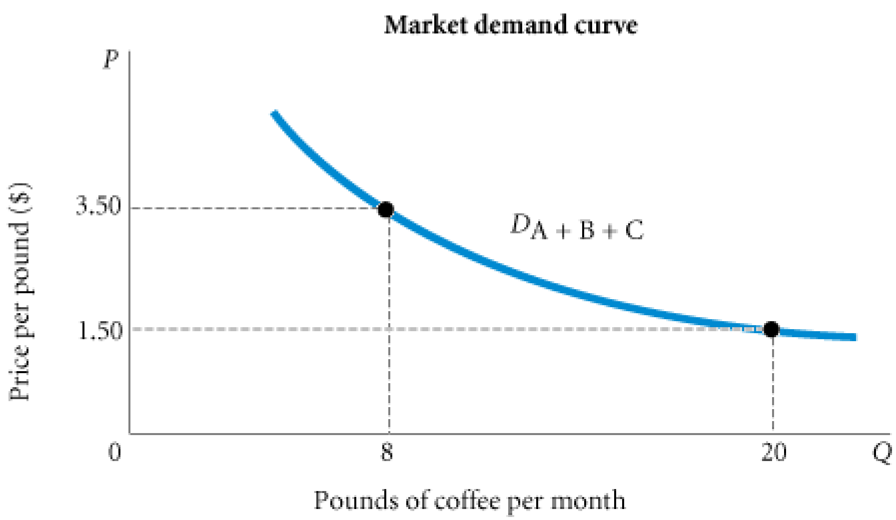
From Household Demand To Market Demand

market demand The sum of all the quantities of a good or service demanded per period by all the households buying in the market for that good or service.

Demand in Product/Output Markets

From Household Demand To Market Demand





Price	Quantity (q) demanded by A	B	C	Total quantity demanded in the market (Q)
$3.50	4 +	0 +	4	= 8
1.50	8 +	3 +	9	= 20

 FIGURE 3.5 Deriving Market Demand from Individual Demand Curves

Total demand in the marketplace is simply the sum of the demands of all the households shopping in a particular market. It is the sum of all the individual demand curves—that is, the sum of all the individual quantities demanded at each price.

Supply in Product/Output Markets

Successful firms make profits because they are able to sell their products for more than it costs to produce them.

profit The difference between revenues and costs.

Supply in Product/Output Markets

Price and Quantity Supplied: The Law of Supply

quantity supplied The amount of a particular product that a firm would be willing and able to offer for sale at a particular price during a given time period.

supply schedule A table showing how much of a product firms will sell at different prices.

Supply in Product/Output Markets

Price and Quantity Supplied: The Law of Supply

law of supply The positive relationship between price and quantity of a good supplied: An increase in market price will lead to an increase in quantity supplied, and a decrease in market price will lead to a decrease in quantity supplied.

supply curve A graph illustrating how much of a product a firm will sell at different prices.

Supply in Product/Output Markets

Price and Quantity Supplied: The Law of Supply

TABLE 3.3 Clarence Brown's Supply Schedule for Soybeans

Price (Per Bushel)	Quantity Supplied (Bushels Per Year)
$1.50	0
1.75	10,000
2.25	20,000
3.00	30,000
4.00	45,000
5.00	45,000

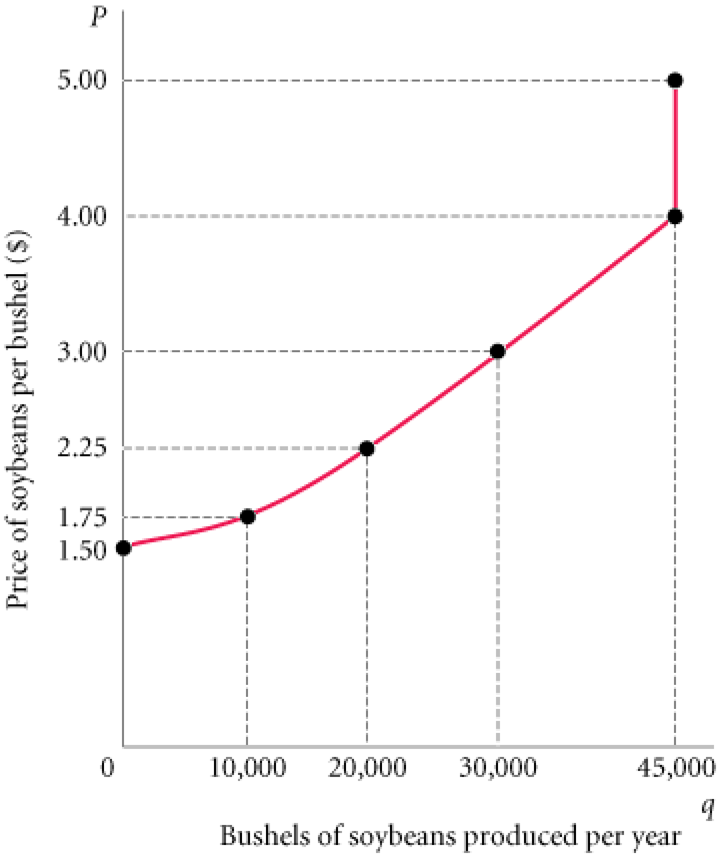


FIGURE 3.6 Clarence Brown's Individual Supply Curve

A producer will supply more when the price of output is higher. The slope of a supply curve is positive. Note that the supply curve is red: Supply is determined by choices made by firms.

Other Determinants Of Supply

The Cost Of Production

In order for a firm to make a profit, its revenue must exceed its costs.

Cost of production depends on a number of factors, including the available technologies and the prices and quantities of the inputs needed by the firm (labor, land, capital, energy, and so on).

Supply in Product/Output Markets

Other Determinants Of Supply

The Prices of Related Products

Assuming that its objective is to maximize profits, a firm's decision about what quantity of output, or product, to supply depends on:

1. The price of the good or service.
2. The cost of producing the product, which in turn depends on:
   - The price of required inputs (labor, capital, and land).
   - The technologies that can be used to produce the product.
3. The prices of related products.

Supply in Product/Output Markets

Shift of Supply versus Movement Along a Supply Curve

movement along a supply curve The change in quantity supplied brought about by a change in price.

shift of a supply curve The change that takes place in a supply curve corresponding to a new relationship between quantity supplied of a good and the price of that good. The shift is brought about by a change in the original conditions.

Supply in Product/Output Markets

Shift of Supply versus Movement Along a Supply Curve

TABLE 3.4 Shift of Supply Schedule for Soybeans Following Development of a New Disease-Resistant Seed Strain

	SCHEDULE D_0	SCHEDULE D_1
Price (per Bushel)	**Quantity Supplied (Bushels per Year Using Old Seed)**	**Quantity Supplied (Bushels per Year Using New Seed)**
$1.50	0	5,000
1.75	10,000	23,000
2.25	20,000	33,000
3.00	30,000	40,000
4.00	45,000	54,000
5.00	45,000	54,000

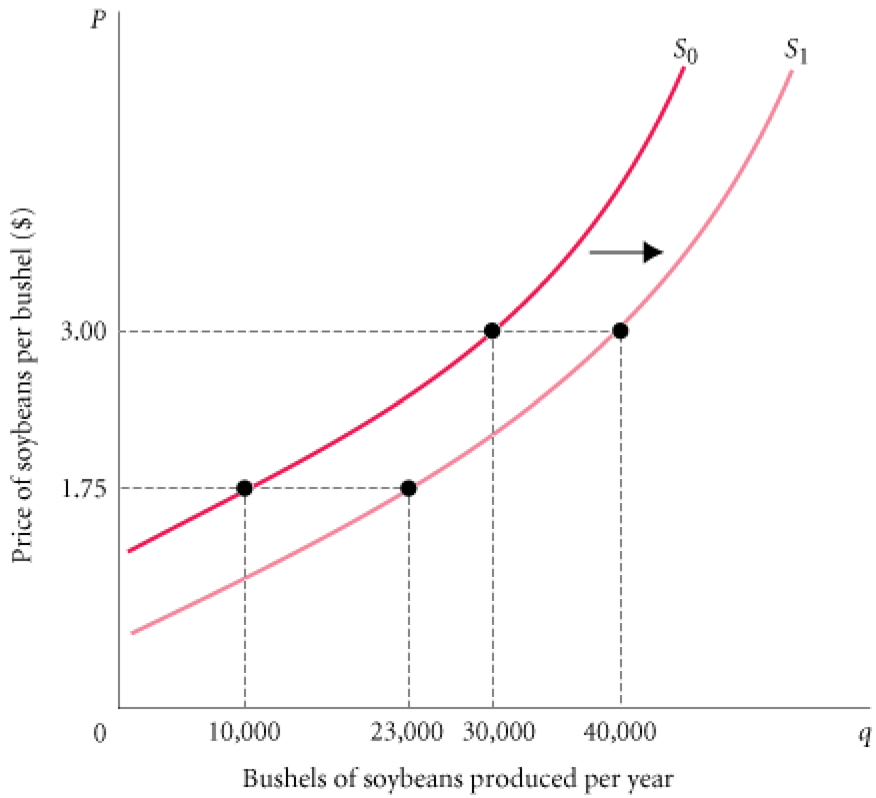


FIGURE 3.7 Shift of the Supply Curve or Soybeans Following Development of a New Seed Strain

When the price of a product changes, we move *along* the supply curve for that product; the quantity supplied rises or falls.
When any other factor affecting supply changes, the supply curve *shifts*.

Supply in Product/Output Markets

Shift of Supply versus Movement Along a Supply Curve

As with demand, it is very important to distinguish between *movements along* supply curves (changes in quantity supplied) and *shifts* in supply curves (changes in supply):

Change in price of a good or service leads to
→ Change in *quantity supplied* (movement along a supply curve).

Change in income, preferences, or prices of other goods or services leads to
→ Change in *supply* (shift of a supply curve).

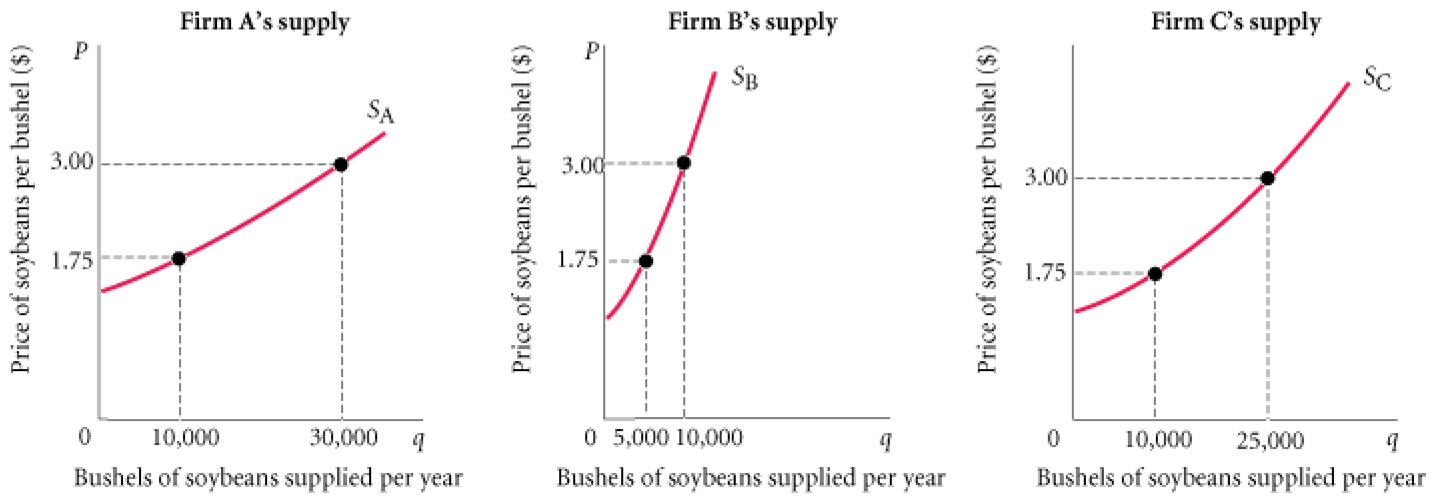
Supply in Product/Output Markets

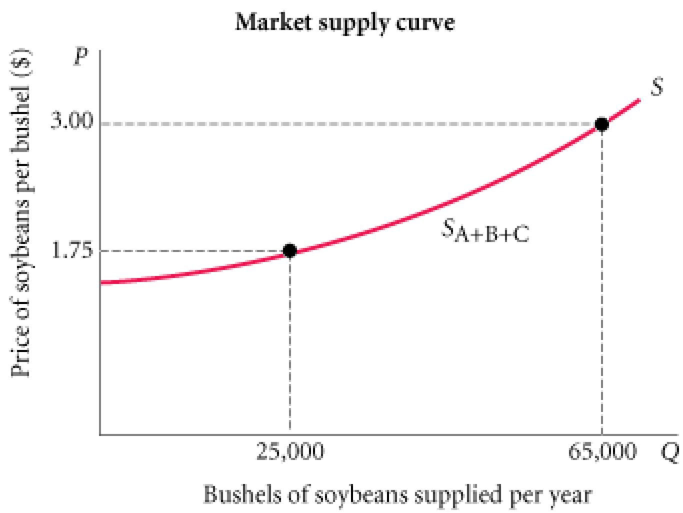
From Individual Supply to Market Supply

market supply The sum of all that is supplied each period by all producers of a single product.

Supply in Product/Output Markets

From Individual Supply to Market Supply





Price	Quantity (*q*) supplied by A	B	C	Total quantity supplied in the market (*Q*)
$3.00	30,000 +	10,000 +	25,000	= 65,000
1.75	10,000 +	5,000 +	10,000	= 25,000

 FIGURE 3.8 Deriving Market Supply from Individual Firm Supply Curves

Total supply in the marketplace is the sum of all the amounts supplied by all the firms selling in the market. It is the sum of all the individual quantities supplied at each price.

Market Equilibrium

equilibrium The condition that exists when quantity supplied and quantity demanded are equal. At equilibrium, there is no tendency for price to change.

Excess Demand

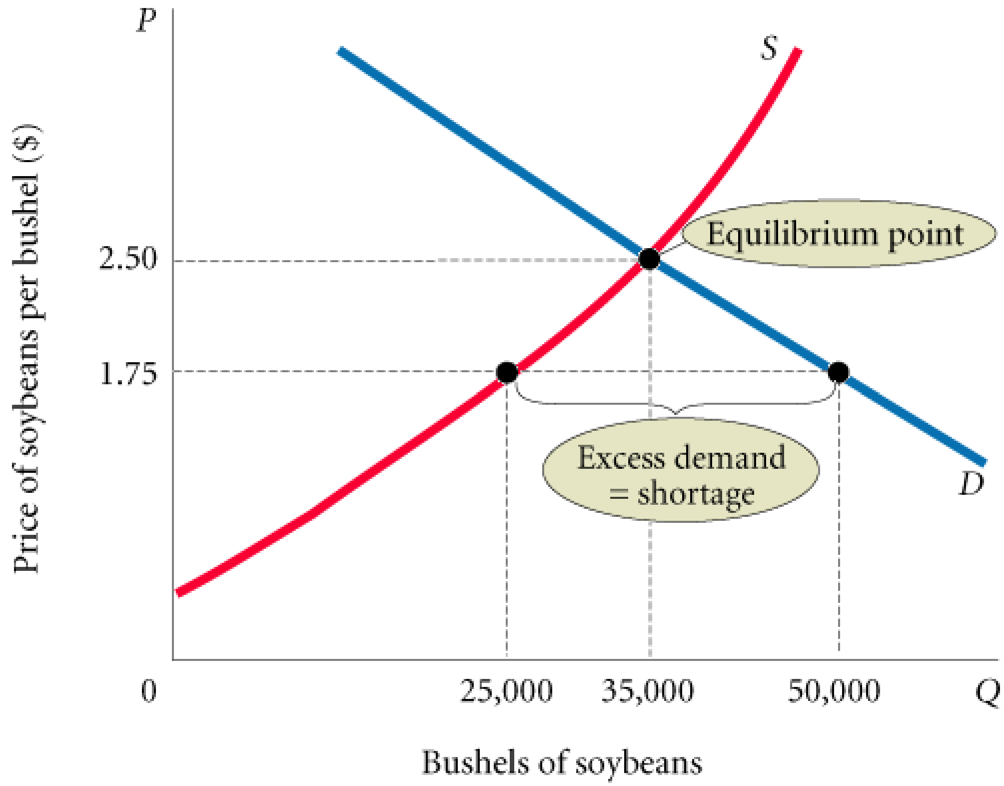
excess demand or shortage The condition that exists when quantity demanded exceeds quantity supplied at the current price.

Market Equilibrium

Excess Demand

 FIGURE 3.9 Excess Demand, or Shortage

At a price of $1.75 per bushel, quantity demanded exceeds quantity supplied. When excess demand exists, there is a tendency for price to rise.
When quantity demanded equals quantity supplied, excess demand is eliminated and the market is in equilibrium. Here the equilibrium price is $2.50 and the equilibrium quantity is 35,000 bushels.



When quantity demanded exceeds quantity supplied, price tends to rise. When the price in a market rises, quantity demanded falls and quantity supplied rises until an equilibrium is reached at which quantity demanded and quantity supplied are equal.

Market Equilibrium

Excess Supply

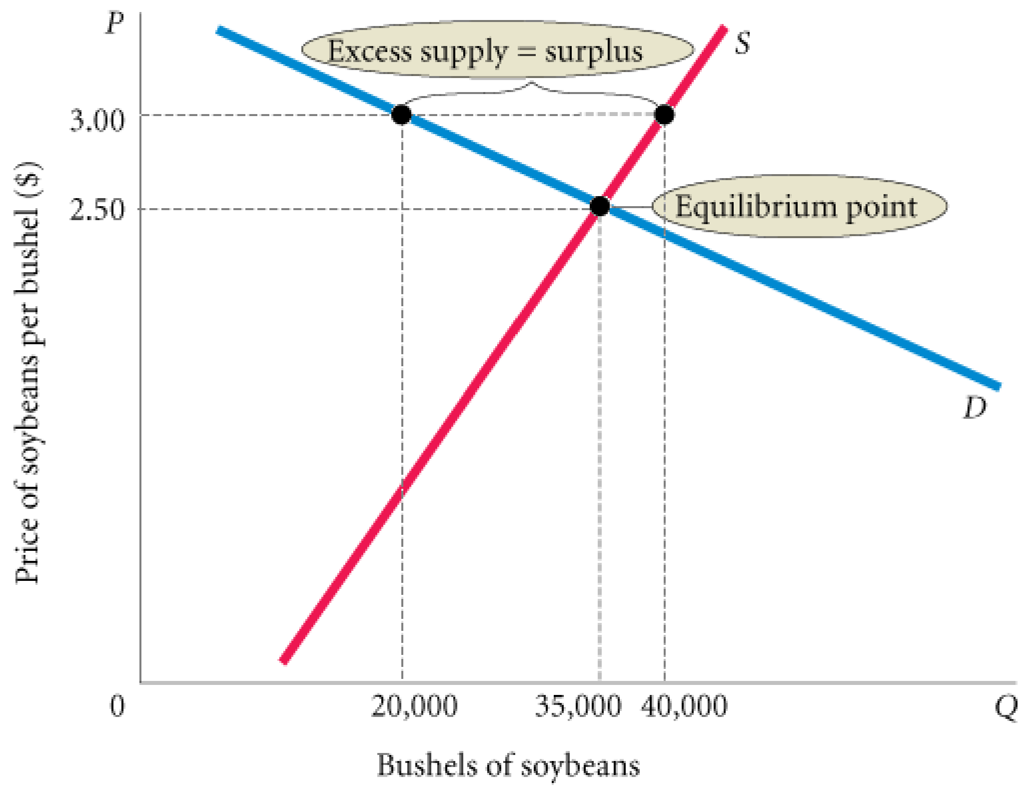
excess supply *or* surplus The condition that exists when quantity supplied exceeds quantity demanded at the current price.

Market Equilibrium

Excess Supply

 FIGURE 3.10 Excess Supply, or Surplus

At a price of $3.00, quantity supplied exceeds quantity demanded by 20,000 bushels.
This excess supply will cause the price to fall.



When quantity supplied exceeds quantity demanded at the current price, the price tends to fall. When price falls, quantity supplied is likely to decrease and quantity demanded is likely to increase until an equilibrium price is reached where quantity supplied and quantity demanded are equal.

Market Equilibrium

Changes In Equilibrium

When supply and demand curves shift, the equilibrium price and quantity change.

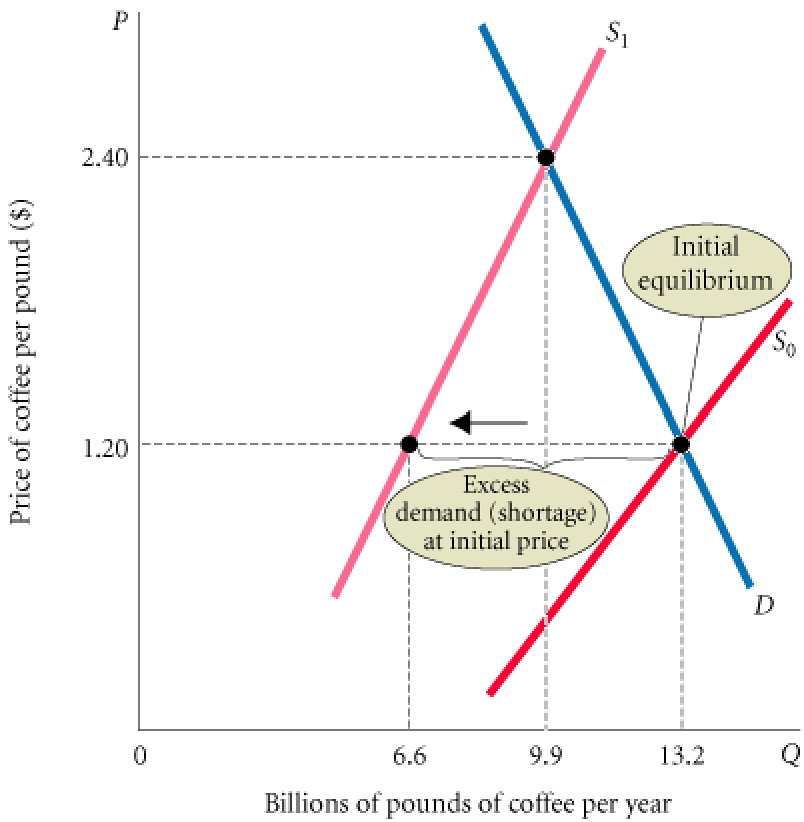


FIGURE 3.11 The Coffee Market: A Shift of Supply and Subsequent Price Adjustment

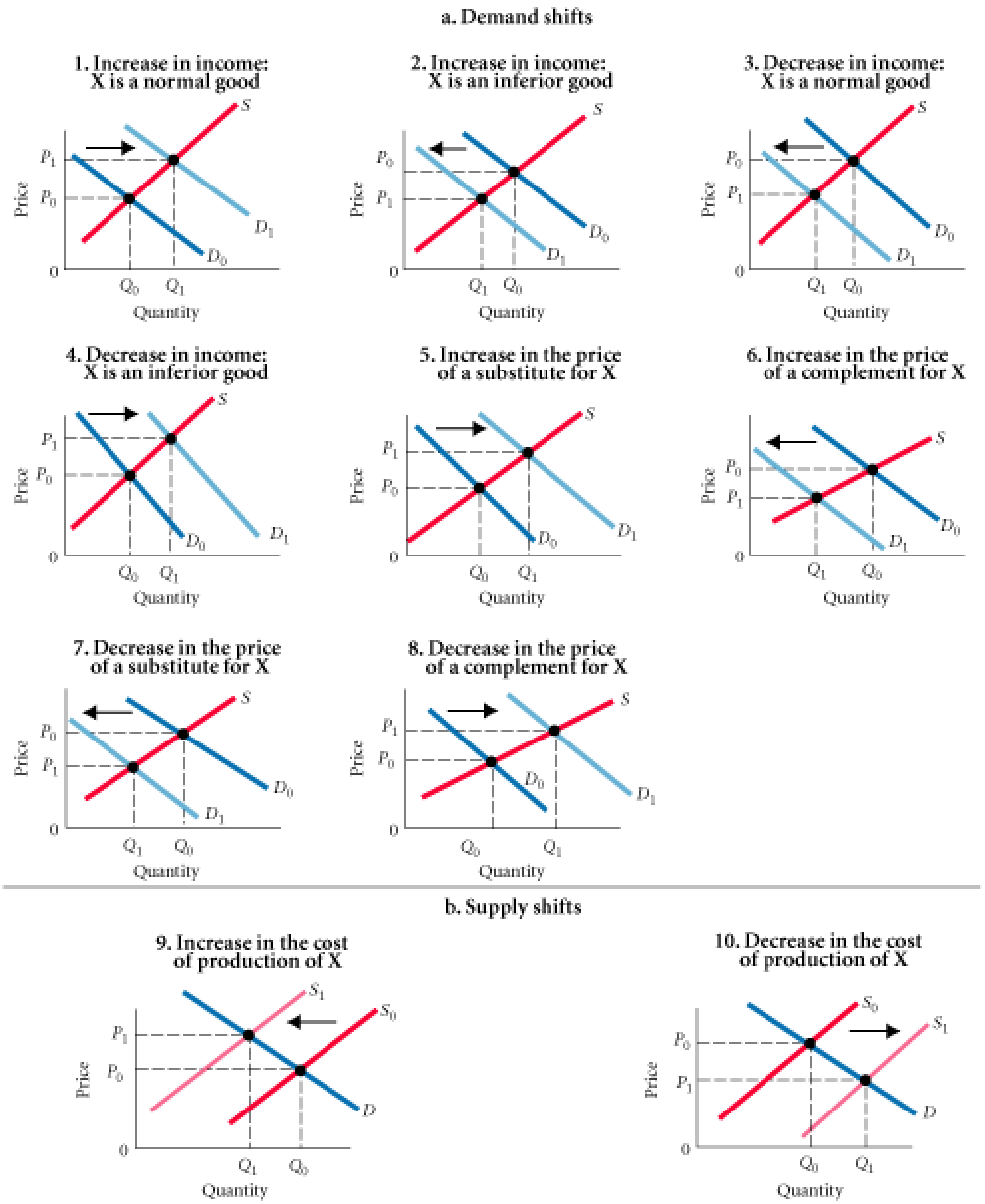
Before the freeze, the coffee market was in equilibrium at a price of $1.20 per pound. At that price, quantity demanded equaled quantity supplied.

The freeze shifted the supply curve to the left (from S_0 to S_1), increasing the equilibrium price to $2.40.

Market Equilibrium

Changes In Equilibrium

 FIGURE 3.12 Examples of Supply and Demand Shifts for Product X



Market Equilibrium

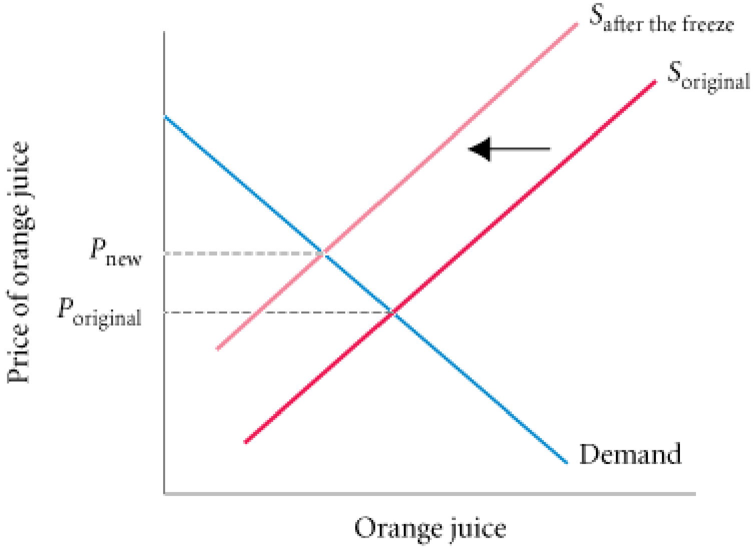
Changes In Equilibrium

ECONOMICS IN PRACTICE

Bad News for Orange Juice Fanatics

Orange Juice Prices Could Skyrocket After Freeze Destroys Most of California Output

City News



Demand and Supply in Product Markets: A Review

Here are some important points to remember about the mechanics of supply and demand in product markets:

1. A demand curve shows how much of a product a household would buy if it could buy all it wanted at the given price. A supply curve shows how much of a product a firm would supply if it could sell all it wanted at the given price.
2. Quantity demanded and quantity supplied are always per time period—that is, per day, per month, or per year.
3. The demand for a good is determined by price, household income and wealth, prices of other goods and services, tastes and preferences, and expectations.

Demand and Supply in Product Markets: A Review

Here are some important points to remember about the mechanics of supply and demand in product markets:

4. The supply of a good is determined by price, costs of production, and prices of related products. Costs of production are determined by available technologies of production and input prices.

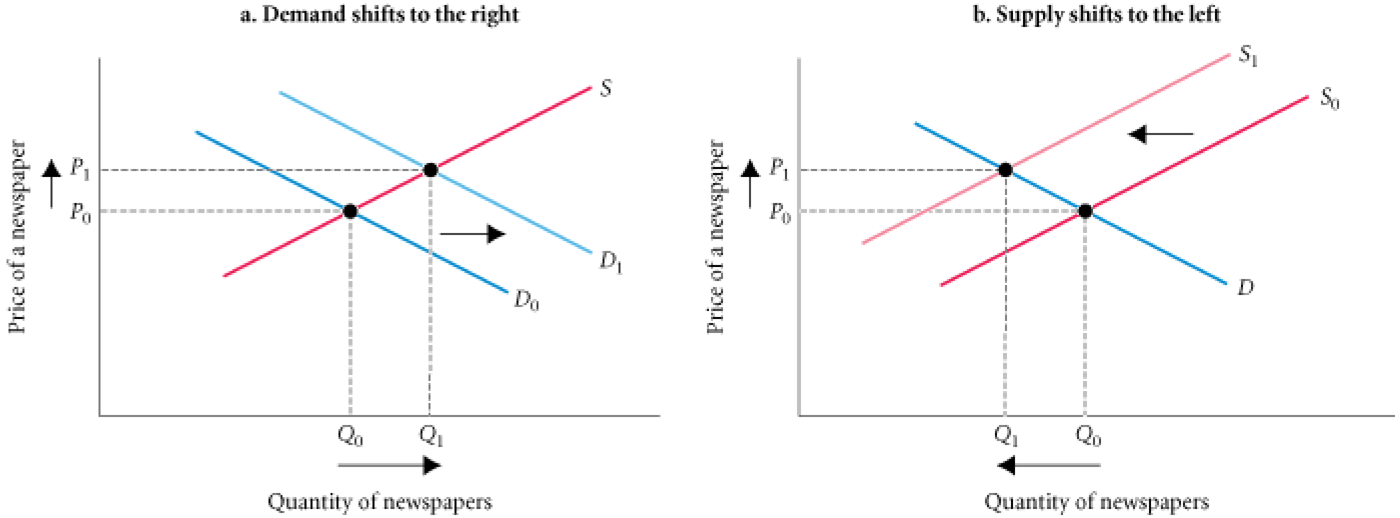
5. Be careful to distinguish between movements along supply and demand curves and shifts of these curves. When the price of a good changes, the quantity of that good demanded or supplied changes—that is, a movement occurs along the curve. When any other factor changes, the curve shifts, or changes position.

6. Market equilibrium exists only when quantity supplied equals quantity demanded at the current price.

ECONOMICS IN PRACTICE

Why Do the Prices of Newspapers Rise?

In 2006, the average price for a daily edition of a Baltimore newspaper was $0.50. In 2007, the average price had risen to $0.75.



Looking Ahead: Markets and the Allocation of Resources

You can already begin to see how markets answer the basic economic questions of what is produced, how it is produced, and who gets what is produced.

- Demand curves reflect what people are willing and able to pay for products; demand curves are influenced by incomes, wealth, preferences, prices of other goods, and expectations.
- Firms in business to make a profit have a good reason to choose the best available technology—lower costs mean higher profits.
- When a good is in short supply, price rises. As it does, those who are willing and able to continue buying do so; others stop buying.